WORKSHEET

Chapter 6: Roots of complex numbers

1 Find the cube roots of $-8i$ and express your answers in Cartesian form.

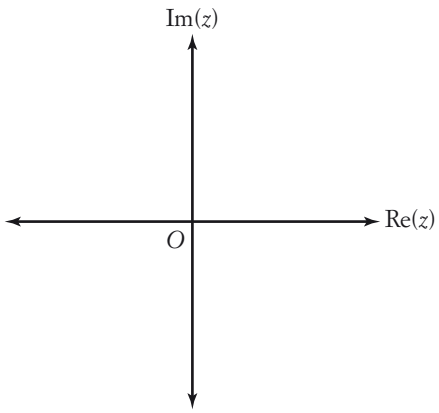
2 Calculate each of the following and express your answers in Cartesian form.

a $\sqrt{-9}$

b $\sqrt{-25}$

c $\sqrt{-18}$

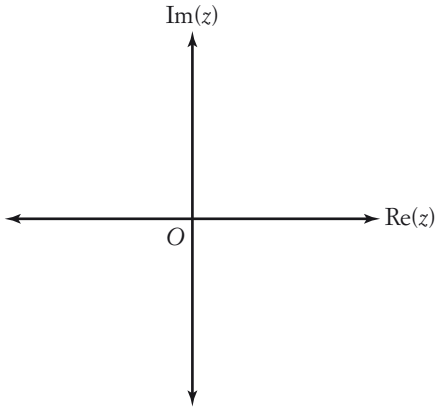
3 Find the sixth roots of $z = -8$ and graph these roots on the complex plane shown here.



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4 **a** Find the cube roots of $(-1 + i)$.

b Show the roots on the complex plane shown here and comment on their positions.



5 Solve $z^3 = i$ and give your answer in Cartesian form.

6 Find the square roots of each of the following and express the answers in Cartesian form.

a i

b $-2i$

c $-3 + 4i$

7 Solve $z^5 + 32 = 0$.

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8 Solve $z^{12} - 1 = 0$.

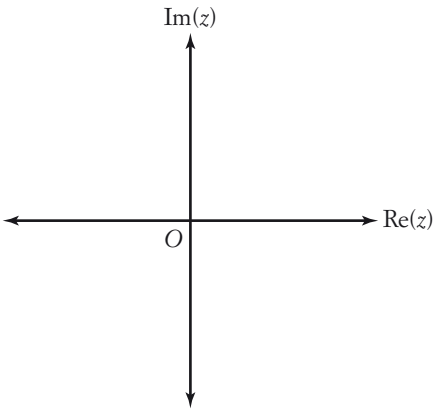
9 Calculate $\sqrt{2i}$.

10 Calculate $\left(\sqrt{3} - i\right)^{\frac{1}{3}}$.

11 Solve the equation $z^2 + 5 = 12i$ and express the answer in Cartesian form.

12 a Find the first two fourth roots of $81[\cos(60°) + i\sin(60°)]$.

b Show the roots on the Argand diagram shown here.



Answers

1 $\sqrt{3}+i, -\sqrt{3}+i, -2$

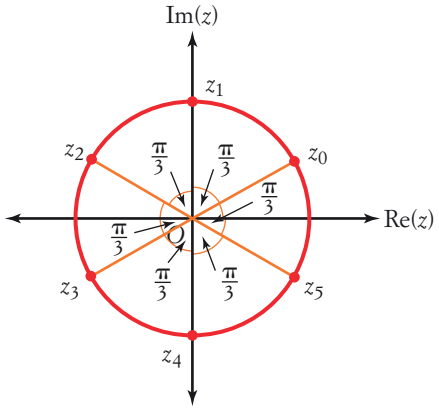
2 **a** $3i$

b $5i$

c $i3\sqrt{2}$

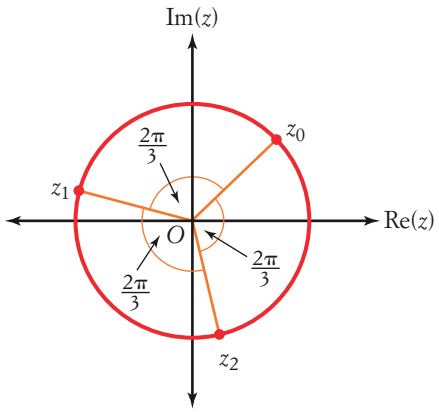
3 $z_0 = \sqrt{2}\left[\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right], z_1 = \sqrt{2}\left[\cos\left(\frac{\pi}{2}\right)+i\sin\left(\frac{\pi}{2}\right)\right], z_3 = \sqrt{2}\left[\cos\left(\frac{5\pi}{6}\right)+i\sin\left(\frac{5\pi}{6}\right)\right],$

$z_3 = \sqrt{2}\left[\cos\left(\frac{7\pi}{6}\right)+i\sin\left(\frac{7\pi}{6}\right)\right], z_4 = \sqrt{2}\left[\cos\left(\frac{3\pi}{2}\right)+i\sin\left(\frac{3\pi}{2}\right)\right], z_5 = \sqrt{2}\left[\cos\left(\frac{11\pi}{6}\right)+i\sin\left(\frac{11\pi}{6}\right)\right]$



The roots equally spaced on a circle of radius $\sqrt{2}$ with centre at the origin.

4 **a** $2^{\frac{1}{6}}\left[\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right], 2^{\frac{1}{6}}\left[\cos\left(\frac{11\pi}{12}\right)+i\sin\left(\frac{11\pi}{12}\right)\right], 2^{\frac{1}{6}}\left[\cos\left(\frac{19\pi}{12}\right)+i\sin\left(\frac{19\pi}{12}\right)\right]$



The roots equally spaced on a circle of radius $2^{\frac{1}{6}}$ with centre at the origin.

5 $\frac{\sqrt{3}}{2}+\frac{i}{2}, \frac{-\sqrt{3}}{2}+\frac{i}{2}, -i$

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6 **a** $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

b $-1 - i$

c $1 + 2i$

7 $z_0 = 2\left[\cos\left(\frac{\pi}{5}\right) + i\sin\left(\frac{\pi}{5}\right)\right], z_1 = 2\left[\cos\left(\frac{3\pi}{5}\right) + i\sin\left(\frac{3\pi}{5}\right)\right], z_3 = -2,$

$z_3 = 2\left[\cos\left(\frac{7\pi}{5}\right) + i\sin\left(\frac{7\pi}{5}\right)\right], z_4 = 2\left[\cos\left(\frac{9\pi}{5}\right) + i\sin\left(\frac{9\pi}{5}\right)\right]$

8 $z = 1, \left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right], \left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right], i, \left[\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right],$

$\left[\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right], \ldots, \left[\cos\left(\frac{11\pi}{6}\right) + i\sin\left(\frac{11\pi}{6}\right)\right]$

9 $1 + i, 1 - i$

10 $\sqrt[3]{2}\left[\cos\left(\frac{-\pi}{18}\right) + i\sin\left(\frac{-\pi}{18}\right)\right], \sqrt[3]{2}\left[\cos\left(\frac{11\pi}{18}\right) + i\sin\left(\frac{11\pi}{18}\right)\right], \sqrt[3]{2}\left[\cos\left(\frac{23\pi}{18}\right) + i\sin\left(\frac{23\pi}{18}\right)\right]$

11 $2 + 3i, -2 - 3i$

12 **a** $z_0 = 3[\cos(15°) + i\sin(15°)] \approx 2.9 + 0.78i, z_1 = 3[\cos(105°) + i\sin(105°)] \approx -7.8 + 2.9i$

b

